# Does Seeing the Search Result Make Tool Utility Predictable?

A diagnostic follow-up to "To Call or Not to Call" (Wu et al., 2026)
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Code: https://github.com/anchalrajput25/tool-utility-reliability

## 1. QUESTION

Wu et al. (2026) find that a probe on an LLM's hidden state predicts whether the model NEEDS a tool well (LNE) but predicts whether the tool will actually HELP (LUE) less well, and attribute this to utility depending on the tool output, which the model has not yet seen. Two other explanations are possible: (a) the utility labels are noisy, since each comes from one no-tool run and one tool run scored by a single LLM-judge call; (b) the information-bottleneck explanation is correct. We test both. Hypothesis: if (b) holds, a probe that sees the retrieved results should predict utility much better, ideally approaching the agreement level of the labels themselves.

## 2. SETUP

Model: Llama-3.2-3B-Instruct (ungated unsloth copy of the same weights), vLLM, temperature 0, the authors' harness unmodified (base local loop). Data: BFCL web-search split, 314 questions; conditions no tool, and always-search with Google (SerpAPI, top-5 snippets, Oct 2026). Judge: the paper's exact BFCL judge prompt on Qwen2.5-7B-Instruct (4-bit, temperature 1.0) instead of GPT-4o; every answer judged K=3 times. Labels: need = no-tool answer wrong; utility = tool score > no-tool score. Probes: the paper's MLP grid on standardized final-layer last-token states. PRE = no-tool prompt (what LUE_x sees); POST = final prompt containing the retrieved snippets. Nested CV: hyperparameters selected inside each training fold; outer 5-fold stratified out-of-fold predictions, identical folds for PRE and POST, seed 42. Statistics: Fleiss kappa and flip rate over judge repeats; bootstrap 95% CIs over questions; paired bootstrap for the PRE-vs-POST AUROC difference.

## 3. DATA QUALITY CHECKS (reported before results)

Our always-search accuracy (0.40 on all 314) was well below the paper's 0.65. We traced this rather than assume the judge was responsible. Time-sensitive questions are NOT the cause (0.41 on "current/latest", 0.37 fixed-year, 0.41 stable). In 60/314 runs (19%) the model called the search tool with a wrong argument name and the harness returned an error string instead of results (accuracy 0.08); NOTE: these runs use the base local loop, the repo's recommended path for vLLM, which differs from the chat-template harness the paper uses for Llama-3.2-3B, so this harness choice may itself produce the argument errors (a chat-harness re-run is a priority). 22 questions are context-dependent ("Where is his hometown?", accuracy 0.09); 2 searches failed on API quota. We report the main analysis on the CLEAN set (232 questions where search ran and the question is standalone; 229 with hidden states): an error string in the prompt is an easy cue that search did not help and would inflate the POST probe. On the clean set, accuracy is 0.12 without the tool and 0.50 with Google (paper 0.17 / 0.65). The remaining gap is consistent with retrieval quality: the gold answer appears in the snippets for 47% of questions (0.74 when it does, 0.09 when not; the second group includes the 60 runs that received no results).

## 4. RESULTS (clean set)

Q1 - Are the labels reliable? Utility labels: Fleiss kappa 0.97 [0.94, 0.99]; full-set per-question flip rate 0.03 [0.01, 0.05], need 0.01; leave-one-repeat-out judge-agreement AUROC 0.99. Repeated judgments are highly consistent, so repeat-to-repeat disagreement alone is unlikely to explain the gap. This measures judge consistency. For correctness, an independent second model (Claude) labelled the 48 of 50 answers it could resolve blind to the judge (ids 23 and 37 truncated in logging and excluded; stratified, not representative): agreement 44/48 = 92%, Cohen's kappa 0.83. 3 of 4 disagreements were the judge marking a correct answer wrong, i.e. the judge errs on the strict side.

Q2 - Does seeing the retrieved results make utility predictable? @@TABLE@@

Seeing the evidence improves utility prediction; relative to the judge-agreement reference (0.99) it closes about 37% of the PRE probe's distance (secondary, descriptive). The full uncleaned set gives the same direction (0.68 -> 0.82, +0.14 [0.07, 0.21]).

## 5. WHAT THIS MEANS

Repeated judgments were highly consistent (~97% agreement), so repeat-to-repeat label disagreement alone is unlikely to explain why utility is hard to predict. Access to the tool output carries real signal (+0.15 AUROC), supporting the paper's explanation in part. But substantial predictive difficulty remains (0.75). Competing explanations we have NOT separated: retrieval quality (the evidence often lacks the answer), whether the model actually uses retrieved evidence, answer-generation errors, and limits of a last-token probe. In particular, the POST prompt is longer and differently formatted than the PRE prompt, so part of the +0.15 may reflect format rather than evidence content; a mismatched-evidence control (src/control_experiments.py) is designed to separate these. The POST probe is a diagnostic, not a deployable pre-call controller: the search cost has already been paid.

## 6. DIFFERENCES FROM THE ORIGINAL SETUP

Base local loop rather than the chat-template harness the paper uses for Llama-3.2-3B (may itself cause the tool-argument errors and the lower always-search accuracy; a chat-harness re-run is the top priority); Qwen2.5-7B judge instead of GPT-4o; ungated copy of Llama-3.2-3B; searches run Oct 2026 (snippets differ); temperature 0 (repo default 1.0); Google only; 3B model only.

## 7. LIMITATIONS AND NEXT STEPS

Judge correctness checked against a second model, not humans (92%, kappa 0.83 on 48 of 50 answers; 2 truncated and excluded); a human-labelled sample would be stronger. One model, one task, one engine; clean-set filtering is post hoc (full-set reported alongside). Next: (i) separate "answer present in evidence" from "model uses evidence" by conditioning the POST probe on evidence hit/miss; (ii) a second search engine to measure search-side label noise; (iii) larger models and the paper's other

tasks; (iv) the content-vs-format controls in src/control_experiments.py (see EXPERIMENTS.md).

References: Wu et al. 2026, To Call or Not to Call, arXiv:2605.00737. Asai et al. 2024, Self-RAG. Yan et al. 2024, Corrective RAG.